

Multimodal and Hybrid Image Analysis of Synthetic Fluorescence Microscopy Images

1. Methods and Task Results

A synthetic nuclei dataset (80 train, 20 validation, 12 unseen test images) was processed as 256×256 grayscale images. The workflow combined direct multimodal description, classical Otsu segmentation, U-Net segmentation and a structured LLM interpretation pipeline.

Task 1 — Direct multimodal description

The representative image was selected by mean-intensity proximity to the dataset mean. A naive prompt produced an unsupported “electron microscope” interpretation, whereas the constrained prompt returned fluorescence microscopy, tissue_type = “uncertain”, visible features and image-quality fields. Llama 3.2 Vision was attempted but failed locally with the Ollama mllama architecture error; Qwen2.5-VL 3B was therefore used with the same procedure.

Optimised Task 1 : Describe only visible evidence; do not diagnose or invent tissue, organ, stain, species or acquisition method. If unsupported, return “uncertain”. Return valid JSON with modality, tissue_type, notable_features and image_quality {overall, contrast, noise, issues}.

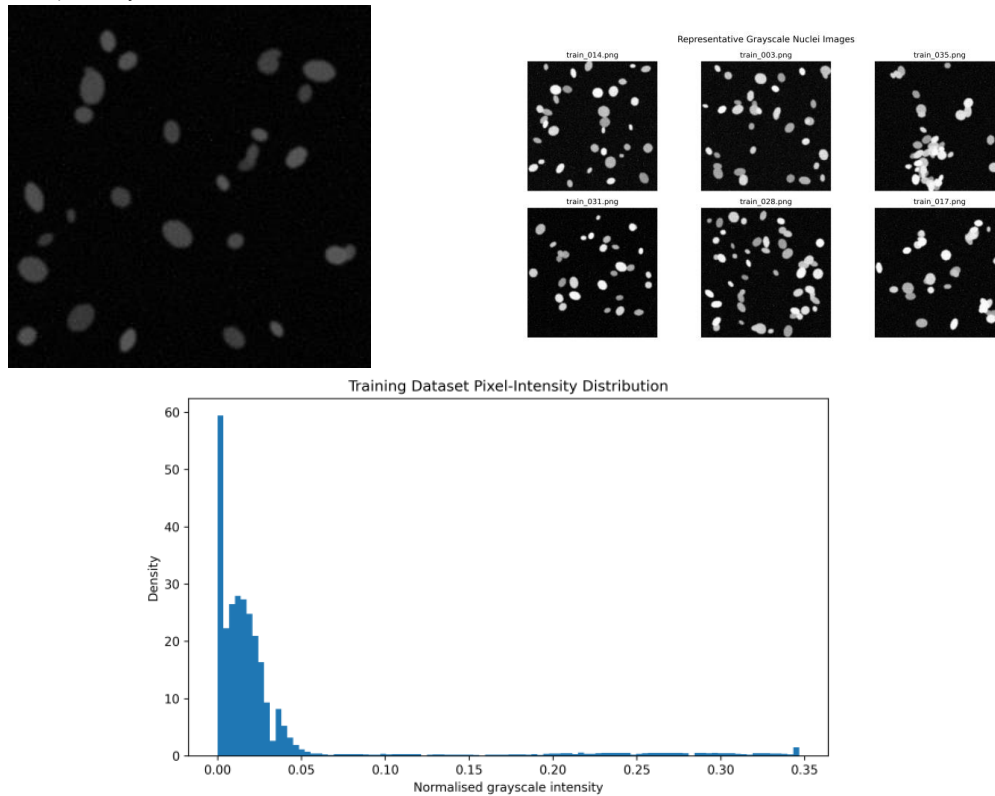


Figure 1. Representative VLM image, sample grayscale images and training intensity distribution.

Task 2 — Classical segmentation and numbers-first LLM

Otsu thresholding, morphology and connected-component labelling were followed by regionprops_table. The representative image contained 24 objects (mean area 172.75 px, eccentricity 0.703, solidity 0.955, foreground fraction 0.063). Only these deterministic measurements were passed to the LLM, making the numbers-first description auditable.

Optimised Task 2 : Use only supplied numerical measurements; do not invent tissue, disease, organ, cell type, stain, species or acquisition method. Return valid JSON with description, n_objects, density_class, shape_regularity and quality_flag; Python-derived values must not be changed.

Task 3 — U-Net segmentation

A small PyTorch U-Net (483,153 trainable parameters) was trained for 15 epochs with Adam and BCE + Dice loss. Evaluation used the untouched 20-image validation set.

Method	Mean Dice	Mean IoU
Otsu	0.9782 ± 0.0027	0.9574 ± 0.0052
U-Net	0.9960	0.9921
Improvement	+0.0178	+0.0348

U-Net achieved higher Dice on all 20 validation images. Otsu remained a strong baseline because foreground and background were well separated, but U-Net better preserved spatial boundaries.

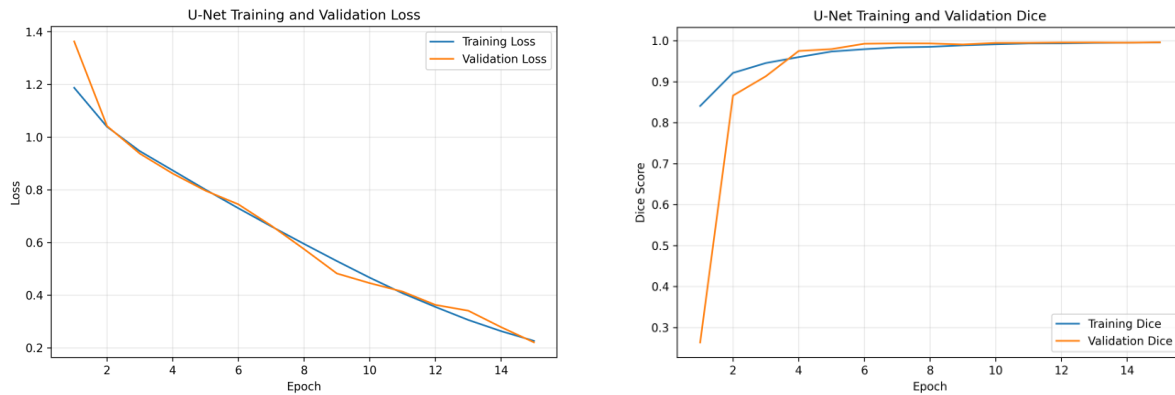


Figure 2a. Training/validation loss and Dice curves.

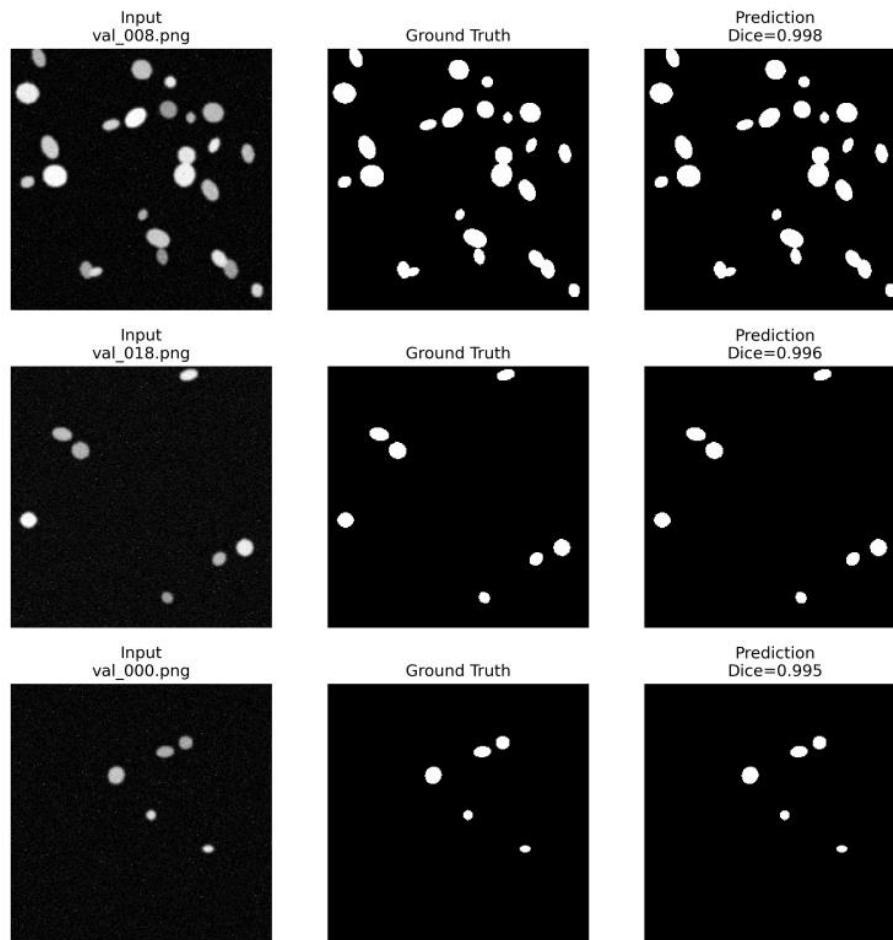


Figure 2b. Input | ground truth | U-Net prediction on three validation images.

Extra credit — loss ablation: BCE, Dice and BCE+Dice were compared in matched short runs. BCE achieved the highest validation Dice (0.9954) for this dataset/configuration.

Task 4 — Hybrid pipeline

All 12 unseen test images were processed as: image → U-Net mask → connected components → regionprops → deterministic source record → constrained LLM JSON → narrative → DataFrame/CSV. Python computed image_id, n_objects, mean_area, density_class and quality_flag; the LLM was not allowed to change them. Structured-output validation passed for 100% of test records.

Optimised Task 4 : Treat the supplied U-Net/Python record as the source of truth. Return JSON containing exactly image_id, n_objects, mean_area, density_class, quality_flag and one concise narrative. Do not change source fields or invent tissue, disease, organ, stain, species or biological identity.

Mean objects 20.67	Mean area 254.83 px	Density split 6 low / 6 moderate
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2. Critical Analysis and Required Questions

1. Which description was more useful and which was more trustworthy?

The direct VLM was more useful for visual appearance, but the numbers-first description was more trustworthy and auditable. The naive VLM hallucinated electron microscopy and possible microorganisms, while the numbers-first LLM was anchored to Python measurements. The optimised VLM improved reliability by permitting “uncertain”.

2. Did U-Net improve on Otsu?

Yes. Mean Dice increased from 0.9782 to 0.9960 and mean IoU from 0.9574 to 0.9921. U-Net was better on every validation image. Otsu remained strong on high-contrast synthetic images, but a single global threshold is less flexible when intensity or object boundaries vary.

3. What do the Dice and IoU values mean, and where did errors occur?

Dice and IoU measure overlap between predicted and ground-truth masks; values near 1 indicate strong agreement. Dice 0.9960 and IoU 0.9921 therefore show near-complete overlap on this validation set. Remaining errors were mainly small boundary and low-intensity edge differences.

4. Where can the LLM hallucinate, and how was this reduced?

Risk is highest in direct image description and open-ended narrative generation. Mitigations were explicit non-diagnostic constraints, uncertainty language, deterministic numerical source fields, strict JSON schemas, post-generation validation and a deterministic fallback.

5. Would this system be trusted clinically, and what would most improve trustworthiness?

No. Strong synthetic validation performance does not establish clinical generalisation. The single most useful improvement would be independent validation on a large, diverse, expertly annotated real microscopy dataset across laboratories and imaging systems, with uncertainty calibration and human review.

Interpretation note: Density, shape and quality categories were project-defined analytical bins, not clinically validated biological classifications.

3. Extra-Credit Robustness

Heavy Gaussian blur ($\sigma = 4$) was applied to one test image and propagated through the complete pipeline. The clean-versus-blurred U-Net masks gave Dice = 0.7110 and IoU = 0.5516, showing a substantial segmentation change. Feature measurements and the narrative also changed, so the earliest detectable failure occurred at the segmentation stage; the LLM mainly propagated altered upstream evidence.

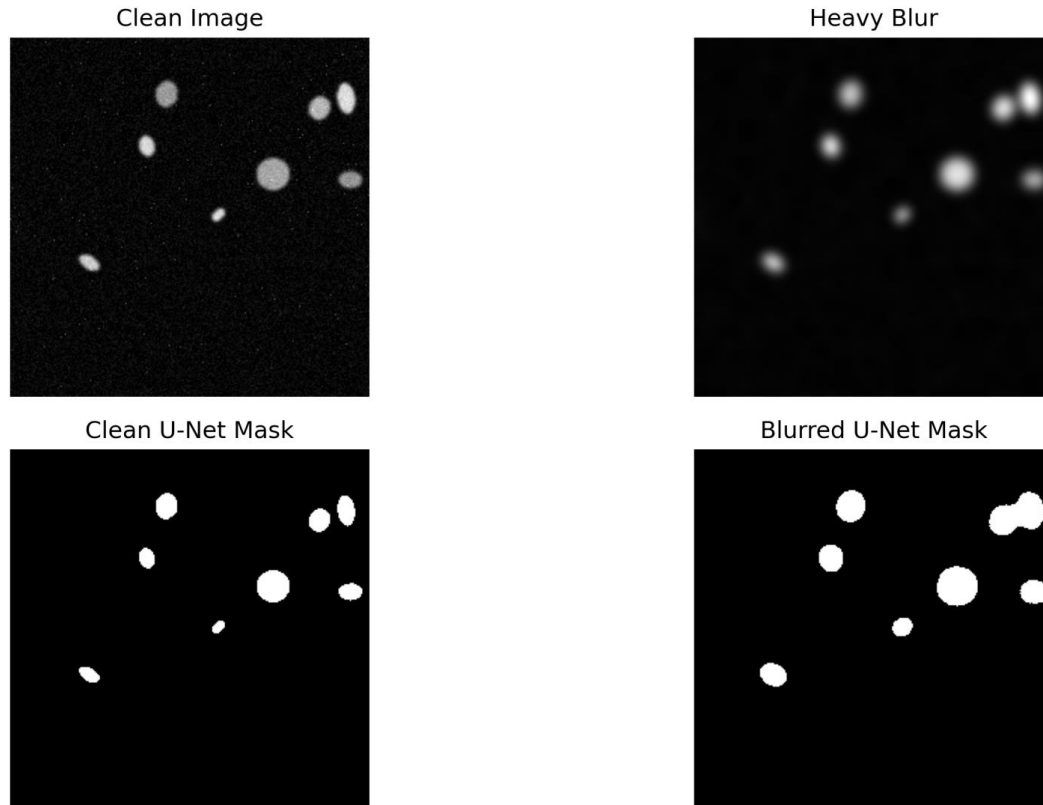


Figure 3. Clean image, heavy blur, clean U-Net mask and blurred U-Net mask.

4. Conclusion

The hybrid pipeline combined interpretable classical measurements with high-accuracy U-Net segmentation and constrained LLM reporting. U-Net clearly improved on Otsu, while numbers-first prompting and JSON validation reduced hallucination risk. The results are promising for a controlled synthetic dataset but are not evidence of clinical readiness.

5. References

1. Otsu, N. (1979). A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), 62–66.
2. Ronneberger, O., Fischer, P. & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI 2015*, 234–241.
3. Korabel, N. (2026). Lecture 3: Multimodal LLMs. Data Analysis with AI, University of Hertfordshire.
4. Bhattacharya, S. (2026). Prompt Engineering, Evaluation & Code Generation with Large Language Models. Data Analysis with AI, University of Hertfordshire.

GitHub Link : <https://github.com/gautam25aas/AI-imaging-Coding-Case-Study-and-Report>